Friction in "Best Yet" Pipes and Fittings Measured by Monometer Pressure Difference

Team: The Boys in the Beaker

Authors: Mark Heien, Nathan Rodrigue, Gabe Wylie, Tommy Yue

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Results

In our experiment, we performed ΔP measurements using a Dwyer Series 490A Hydronic Differential Pressure Manometer across one-meter lengths of pipe on the Armfield C6MK10 Fluid Friction Apparatus. Knowing that pressure in a straight pipe is higher upstream, we respectively attached the "high" and "low" pressure manometer hoses to four smooth steel pipes of inner diameter (ID) 4.5, 7.7, 10.9, and 17.2 mm, and one 15.2 mm ID rough bore, "biomimetic shark skin" pipe. This was done to test the claim that the shark skin pipe was incredibly low friction. We varied flow rate by turning the valve on the water supply, then measured the pressure drop in each pipe for five different flowrates. This gave us several pressure difference values (ΔP) for every pipe at each flowrate (Q). Using this data, we can use Equation 1 to relate Q as a function of ΔP (Edge):

$$Q = A_2 \sqrt{\frac{2}{\rho(1 - \left(\frac{A_2}{A_1}\right)^2}} \sqrt{p_1 - p_2} = B * \sqrt{p_1 - p_2}\,, \qquad 1$$

where B is a lumped constant that is determined by plotting Q against $\sqrt{\Delta P}$ and treating B as the slope of the line, shown in Figure 1.

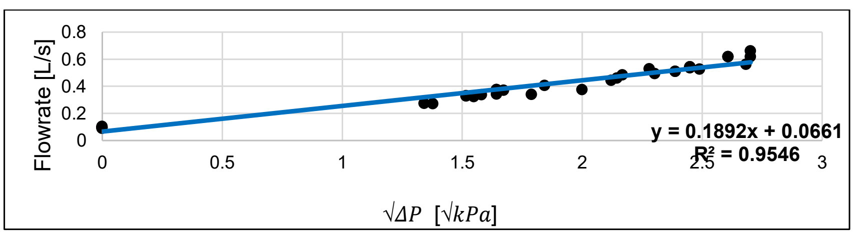

Figure 1. Raw Venturi calibration curve; Q vs. $\sqrt{\Delta P}$

The data shown in Figure 1 describe a linear relationship between flowrate and the square root of ΔP. The slope of the trend-line equation shows that $B = 0.1892$. However, intuitively we expect zero flowrate when the pressure difference is also zero. For this reason, the y-axis intercept of Figure 1 was fixed to (0,0), shown in Figure 2. With this change, the linearity improved compared to the raw calibration curve, as shown by the increase in R^2 values between the two graphs.

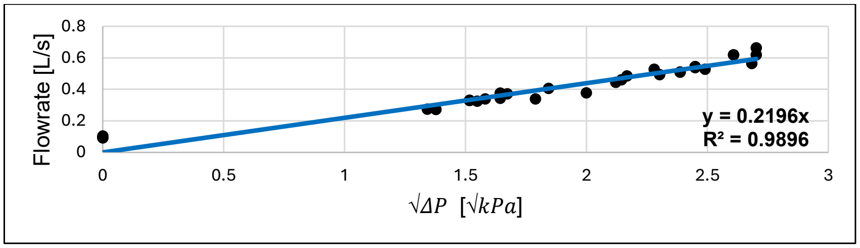

Figure 2. Fixed (0,0) intercept Venturi calibration curve; Q vs. $\sqrt{\Delta P}$

With the intercept fixed at (0,0), we found $B = 0.2196$, and this value could be used for future flowrate measurements without needing to do manual bucket and stopwatch measurements. Using the data analysis add-on in Excel, we calculated a 90% confidence interval of the $\sqrt{\Delta P}$ measurement between 1.53 and 2.03 $\sqrt{kPa}$.

Along with the Venturi meter, the orifice meter was also tested as a means of modeling the flowrate as a function of pressure drop, which can be modeled by the following equation, given constant height.

$$\frac{\Delta P + (\Delta h * \rho g)}{(Q)^2} = \frac{\rho g * ev}{2 * A_C^2} = C = \frac{\Delta P}{Q^2}, \quad 2$$

Following the same procedure as in the venturi meter, the following plots were generated, given free y intercept (figure 3), and set 0 y intercept (figure).

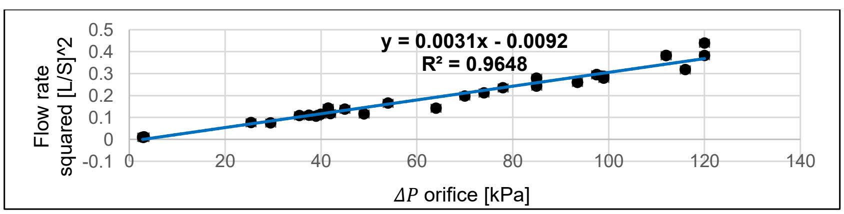

Figure 3. Free y-intercept orifice calibration curve $\Delta P\ vs\ Q^2$

The linear relationship and small intercept in figure 3 show that data fits well to a (0,0) intercept model, and the $R^2 = 0.9648$ shows that the linearity of the data is high. Since this was so close to a 1 variable model, testing a zero intercept model yields figure 4.

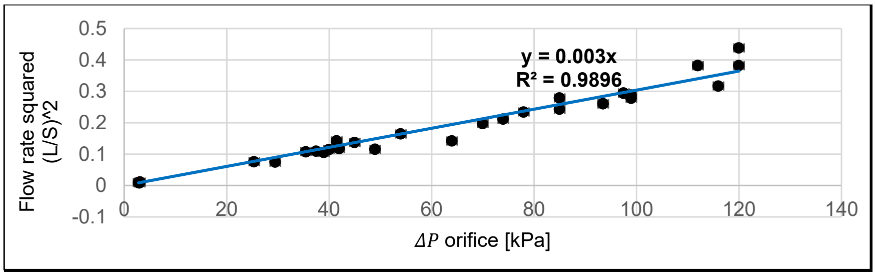

Figure 4. Fixed (0,0) y-intercept orifice calibration curve $\Delta P\ vs\ Q^2$

Figure 4, indicates a better linearity ($R^2 = 0.9896$) than figure 3 ($R^2 = 0.9648$) with (0,0) intercept fixed. The slope value of venturi meter is significantly higher than the slope of the orifice meter, meaning that much more change in pressure will occur per unit of increase in flow rate. This indicates that Venturi has much higher sensitivity and much lower parasitic pressure loss.

Pipe friction scales with both diameter and length of a pipe, for which length constant at 1 meter. Water height in and out of the pipe were held constant, and diameter within each individual pipe was constant. Given measured flow rate and pressure drop, friction factor is assessed by equation 3 (Bird).

$$\frac{\frac{\Delta P}{\rho g} + \Delta h}{2 * L * \langle Q \rangle^2} * A_C^2 * D * g = \frac{\Delta P * D^3 * \pi}{\rho * 2 * 4 * L * \langle Q \rangle^2} = f, \qquad 3$$

Where f=moody friction factor, D=diameter, P=pressure, L= length, Q= volumetric flow rate, and ρ= density of water. Shark skin pipes were evaluated with inner diameter after coating (2 mm less).

Pipe friction factors, reflected in figure 5, indicate that all the stainless-steel pipes are similarly hydraulically smooth, which agrees with theoretical roughness of 0.4-3.2 μm (Threeway Steel Co).

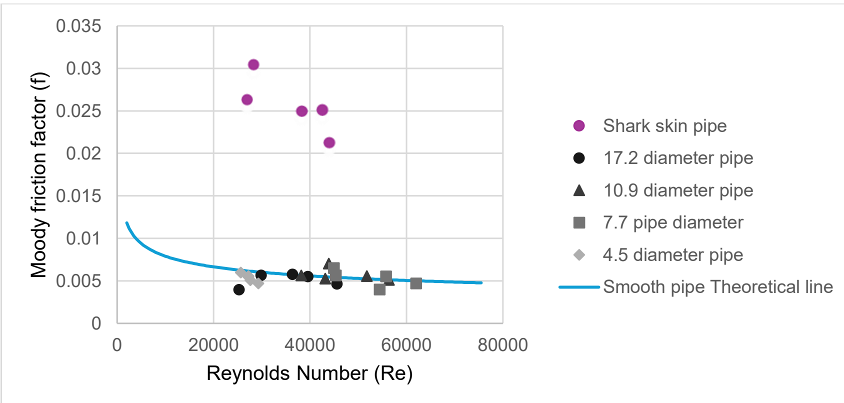

Figure 5. Moody friction factors (f) vs Reynolds Number (Re) of stainless steel and shark pipes

Figure 5 indicates that shark skin pipes are significantly higher friction factor, and therefore frictional losses, than the normal stainless-steel pipe. Shark skin pipes seam to empirically be greater than or equal to friction factors of the roughest pipes normally represented in moody friction charts.

Table 1 indicates our measurements for e_v factor in frictional losses, calculated by equation 4 (Bird). Our results are significantly less than expected, and our first point resulted from 0 change in pressure with height change, indicating value below our ability to measure.

Table 1. Friction e_v experimentally determined vs expected from Chem e 330 material (Adler)

Fitting type	Expected e_v	Experimental e_v
gradual 90 bend		-0.019
45° bend	0.35	0.090
Globe valve	6	0.824
Gate valve	0.17	0.087
90° elbow	0.75	0.216
Tee	1	0.348
90° elbow	0.75	0.180
Union	0.04	0.200

$$ev = \frac{\pi * D^2}{Q^2 * \rho g * 2}(\Delta P + \Delta h * \rho g) \qquad 4$$

Example claculation

$$e_v = \frac{\pi * 17200m^2 * (2600Pa + 998.21\frac{kg}{m^3} * 9.8\frac{m}{s^2} * 0.0508m)}{2 * 0.00062\frac{m^3}{s} * 998.21\frac{kg}{m^3} * 9.8\frac{m}{s^2}} = 0.090$$

Error Analysis

Standard error formula:

$$\% \, Error = \frac{|\, Experimental\ value - Accepted\ value|}{Accetped\ value} \quad 5$$

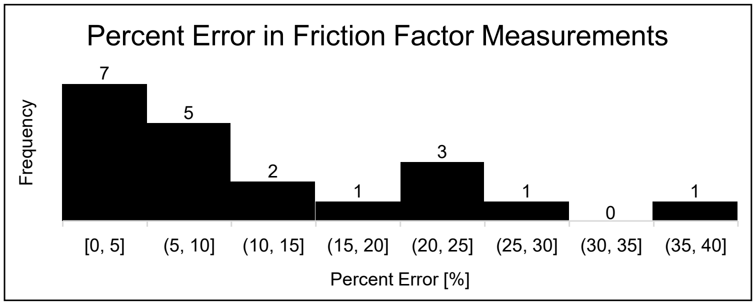

Figure 6. Error% of the smooth pipes except the rough bore pipe.

Our data points are mostly (14/20) acceptable within the 15% error range; it shows that our data and results are reliable to draw conclusions on.

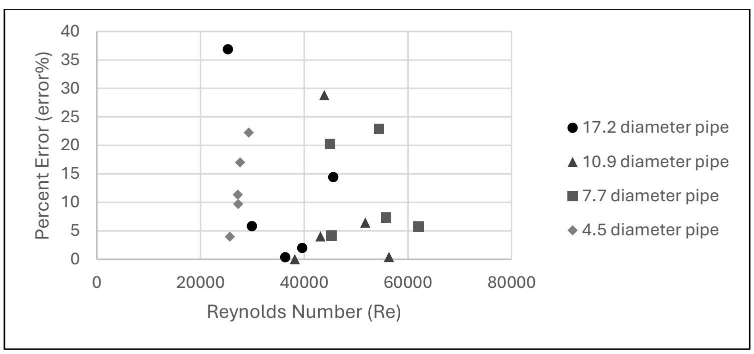

Figure 7. Error% of the smooth pipes except the rough bore pipe.

Two points stand out: ~29% at Re ≈ 41–44k (10.9 mm) and ~37% at Re ≈ 25–30k (17.2 mm). Ranked by typical error: 4.5 mm > 7.7 mm > 10.9 mm ≈ 17.2 mm. Thus, larger pipes provide more stable, lower-error results across the tested range.

One explanation is drops of water were leaking from the Orifice when we were measuring the flowrates and pressures. Meanwhile, the air can go in/out of the pipe which can result in a significant error in our data (biasing Q low and adding ΔP noise), besides the systematic uncertainty.

We also fully connected the manometer and purged the air bubbles in the pipe before every pressure measurement to minimize the error. While the pressure-readings fluctuated every time, we had to approximate a median pressure point +/- 0.3 kPa, which would introduce a random error to our data while also introducing realism. For the slowest flow rate time measurements, we decided to measure only once for time efficiency. Based on the data we collected previously for the faster flowrate which we measured 3 times for each and then averaged the value of time, our measures are close enough for precise data analysis. This may introduce a minor or negligible error on such scale.

Also, we noticed another error during the experiment -- pressure Num of 7.7 diameter pipe 3rd flowrate– when recording the data of 4th flowrate, we noticed an unexpected increase in the pressure. Then we double checked the pressure Num of 4th flowrate is correctly collected, so we conclude that the pressure Num of 3rd flowrate is false. But it's too hard for us to switch the gear back to achieve the exact same flowrate. But that does not cause a noticeable error in the results.

Systematic uncertainty could also include slight variations in pipe diameter and temperature-dependent viscosity of water (assumed constant at 20 °C, 0.0010016 Pa•s). Our manometer has an error of +/-0.5% full scale in room temperature.

In sum, considering all random and systematic contributions, the maximum combined error in friction factor should not be more than 10% in final results which is acceptable for laboratory-scale hydraulic testing.

Discussion

Vendor claims regarding the 15.2mm ("shark skin") pipe, are refuted by our data, which shows a mean f = 0.0256, roughly 5.0 times higher than the 17.2 mm conventional pipe at similar Re (0.00510). Mechanistically, that could arise from either a genuinely rough internal texture (improperly scaled texture for the local boundary layer) or Manufacturing/liner defects or fouling. This data directly contradicts the vendor's claim of "lowest friction ever measured," as it would translate directly into higher head loss and increase energy costs.

The analysis of friction in conventional straight pipes (4.5-17.2 mm) largely validated standard fluid dynamics principles. For the conventional pipes (4.5, 7.7, 10.9, 17.2 mm IDs) operating in the turbulent regime ($2.5 * 10^4 \leq Re \leq 6.2 * 10^4$), the measured Fanning friction factor, (f), clustered around the smooth-pipe Blasius trend ($f = 0.0791Re^{-0.25}$) (Bird). The data for the standard smooth pipes closely followed the established Perry's/Moody chart curve, confirming the reliability of the Boys in the Beaker's experimental setup. The shark skin pipe does not follow this hydraulically smooth pipe trend.

The flow meter calibration fit was very linear on both Venturi and Orifice but were better with no y intercept as follows trends. The Venturi fit $Q = B\sqrt{\Delta P}$ yielded $R^2 \approx 0.99$ similarly the orifice fit $\Delta P = CQ^2$ gave $R^2 \approx 0.99$. Our venturi consistently gave significantly lower pressure drops, so given a flow rate large enough for a monometer to detect the pressure drop, Venturi meters are better than orifice meters. This is consistent with theory; the Venturi's gradual geometry recovers most of the pressure (=90% typical) and disturbs the flow less than an orifice (=60% recovery), so it tends to calibrate" cleaner" and impose less permanent loss. For permanent installations where energy efficiency matters, the Venturi is the better flow meter, given detectable pressure drops.

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Measured value e_v values were low, lending support to the "low loss" valve claim. Given how extraordinary our measurements indicate the e_v to be, it is recommended that values are confirmed with more accurate equipment.

Pump power $\dot{P} \propto \Delta P\ Q$ and ΔP scales with f, adopting the "shark-skin" tube would raise operational expenses; conversely, staying with smooth pipe and Venturi metering reduces lifetime energy draw for a given throughput. Even single-digit percentage reductions in f compound to many-digit energy savings over continuous operation.

Recommendations

The "shark-skin" 15.2 mm tube should be rejected for water service within the tested Reynolds range. Its friction factors were several times higher than the smooth-pipe Blasius correlation, implying greater pressure loss and energy cost. Approval should occur only if the supplier proves, on our rig, a ≥ 15 % lower friction factor across the target range with certified roughness and inner-diameter documentation. Standard smooth commercial steel (or equivalent) piping is recommended for new lines, as it matches theoretical correlations, minimized head loss, and simplified hydraulic design.

Venturi meters should serve as the primary flow-measurement devices. Venturi offers lower permanent losses than orifice plates and can be calibrated to be as good or better than orifice (R^2 = 0.99).

New Fittings and valves offer potentially significant friction advantages over standard valves, but should not be implemented without further testing, since the numbers seem too good to be true, especially the 45 degree and 90-degree elbows.

References

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